Code No: 136BC **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, February - 2023
DESIGN PATTERNS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	What do you mean by design Pattern?	[2]
b)	Write about framework.	[3]
c)	What is Decorator pattern?	[2]
d)	With suitable example, explain about compositor and composition.	[3]
e)	Draw the structure of Prototype pattern.	[2]
f)	What are benefits of Singleton pattern?	[3]
g)	What are related patterns of Façade pattern?	[2]
h)	List the consequences of Composite pattern.	[3]
i)	Write the intent of Observer pattern.	[2]
j)	Describe motivation of Strategy pattern.	[3]

PART – B

(50 Marks)

2.a) What are some common causes of redesign? Explain briefly.
b) How to use a design pattern? Discuss in brief. [5+5]

OR

3. Discuss about design patterns in Smalltalk MVC in detail. [10]

4. Explain in detail about supporting multiple window systems problem in Lexi's design. [10]

OR

5. Describe user operations problem in Lexi's design. [10]

6.a) Write and explain the sample code of Prototype pattern.
b) Describe motivation, applicability and structure of Builder pattern. [5+5]

OR

7.a) Discuss about implementation issues of Abstract Factory pattern.
b) Explain sample code and known issues of Factory Method pattern. [5+5]

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8.a) Describe implementation issues of Composite pattern.
b) Discuss about the implementation of Proxy pattern. [5+5]

OR

9.a) Explain about the motivation and applicability of Flyweight pattern.
b) Write about the sample code of Decorator pattern. [5+5]

10. Explain the following:
a) Sample code of Iterator
b) Template method pattern. [5+5]

OR

11. Write about:
a) Implementation of Chain of Responsibility pattern
b) Sample code of State pattern. [5+5]

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